

Request for Apollo Lunar Thin Sections and Apollo Lunar Soil Samples for Use at NFAR for a Proof-of-concept Study on Automatic Mineral Identification and 3D Reconstructions

1 General Information

Title: Request for Apollo Lunar Thin Sections and Apollo Lunar Soil Samples for Use at NFAR for a Proof-of-concept Study on Automatic Mineral Identification and 3D Reconstructions

Samples to Use:

We request the following 100 Apollo lunar thin sections for the proposed proof-of-concept work:

10003, 10009, 10017, 10018, 10019, 10020, 10021, 10022, 10023, 10024, 10027, 10032, 10044, 10045, 10046, 10047, 10048, 10049, 10050, 10056, 10057, 10058, 10059, 10060, 10061, 10062, 10063, 10064, 10065, 10066, 10067, 10068, 10069, 10070, 10071, 10072, 10073, 10074, 10075, 10082, 12002, 12004, 12005, 12006, 12007, 12009, 12010, 12011, 12012, 12014, 12015, 12016, 12017, 12018, 12019, 12020, 12021, 12022, 12031, 12034, 12035, 12036, 12038, 12040, 12043, 12045, 12047, 12051, 12052, 12053, 12054, 12055, 12062, 12063, 12064, 12065, 12072, 12073, 12075, 12076, 14006, 14035, 14041, 14042, 14045, 14047, 14049, 14051, 14053, 14055, 70002, 20017, 70019, 70135, 70136, 70148, 70161, 70175, 70181, 70185

We also request the following 10 Apollo lunar soil samples:

70011, 70050, 70060, 70161, 70170, 70180, 70181, 70271, 70311, 72221

Users:

We will work with the requested Apollo Thin Sections and Apollo lunar soil at the JSC NFAR; these Apollo thin sections and Apollo lunar soil samples would never leave ARES/JSC during this proposed study.

- Ping Wang, US Citizen, Research Associate, Earth, Environmental, and Planetary Sciences, Department, University of Tennessee, Knoxville, Email: pwang27@utk.edu
- Shichun Huang, US Citizen, Sisk Associate Professor, Earth, Environmental, and Planetary Sciences Department, University of Tennessee, Knoxville, Email: shuang25@utk.edu
- High school students from the East TN area

2 Description of Proposed Research on Apollo Lunar Thin Sections and Apollo Lunar Soil Samples

2.1 Scientific Purpose

The Purpose is twofold-

- a) to create an automatic mineral identification system for Apollo lunar thin sections. We request access to the Apollo lunar thin sections and we will use the requested Apollo lunar thin sections with the petrographic microscopes, scanning electron microscopes (SEM), and technical expertise at NFAR.

- b) to develop 3D reconstructions with the SEM images of the requested Apollo lunar soil samples. We request access to the Apollo lunar soil samples and we will use the requested Apollo lunar soil samples with the scanning electron microscopes (SEM) and technique expertise at NFAR.

We propose to:

- a) generate petrographic images of Apollo lunar thin sections under plane polarized light, cross polarized light, and reflective light with petrographic microscopes
- b) generate elemental maps of studied Apollo lunar thin sections with SEM instruments
- c) build datasets with petrographic images and SEM images for developing a machine learning
- d) generate SEM images of Apollo lunar soil samples for 3D chemical reconstruction
- e) create 3D chemical mapping models based on the SEM images of Apollo lunar soil
- f) develop innovative AI/ML methods for automatic identification in Apollo lunar thin sections and AI methods in mapping geological properties of Apollo lunar thin sections
- g) Develop innovative AI/ML methods to tackle accuracy and uncertainty in mapping of the geochemical properties of Apollo lunar samples.

The goal of this proposal is (1) to gather all petrographic images and elemental maps of Apollo lunar thin sections, which will be used to develop an automatic mineral identification system and AI methods in mapping geological properties of Apollo lunar thin sections; and (2) to gather all SEM images of Apollo lunar soil for 3D reconstruction for chemical mapping.

We expect to collaborate with **Dr. Justin Simon** at ARES to complete this proof-of-concept project. If successful, we plan to submit a proposal to NASA The Lunar Data Analysis Program (LDAP) to further our effort developing an automatic mineral identification system. This collaboration would continue the long history of lunar exploration study at the University of Tennessee, Knoxville.

All the petrographic images and elemental maps of Apollo lunar thin sections, and SEM images of Apollo lunar soil samples produced in this study can be used independently for future research, education, and outreach activities. For example, we plan to build a virtual microscope system to make the petrographic images and elemental maps of Apollo lunar thin sections, 3D chemical mapping models of Apollo lunar soil produced in this study available to the public.

2.2 Our Approach

Our ultimate goal is to develop AI/ML algorithms to automatically identify minerals in natural rocks, and also develop innovative AI/ML methods to tackle accuracy and uncertainty during 3D Reconstructions for 3D chemical mapping for lunar samples.

As one sub-task, in this proof-of-concept study, we aim to develop AI/ML methods that allow automatic identification of minerals in rock thin sections of lunar rocks returned by Apollo missions. In this approach, we will use chemical analysis by SEM to independently label each

mineral grain in our studied thin sections. These labeled minerals will serve as the ground truth in developing AI algorithms.

Optical Method: Minerals can be identified using a variety of methods. A traditional method for identifying and classifying minerals is to exam thin sections under petrographic microscope, and minerals are identified based on their optical properties, such as birefringence [1]. The use of a petrographic microscope for mineral identification requires highly trained personnel. Mineral identification is possible using the specific polarized transmitted and reflected light properties of minerals and the morphological properties of the grains. Most minerals can be identified when examined with a petrographic microscope, even if unidentifiable in hand specimens. Optical properties allow us to estimate the composition of some minerals. We use petrographic microscopes with orthoscopic illumination, and with, or without, the upper polarizer inserted. Without the upper polarizer, we observe a sample in plane-polarized light, also called PP light. With the upper polarizer, we observe it in cross-polarized light, also called crossed polars or XP light. Grain size, shape, color, cleavage, and some physical properties are best revealed in PP light. The optical properties of the refractive index and pleochroism are also determined using PP light. We use XP light, sometimes focused with conoscopic and Bertrand lenses, to determine other properties including retardation, optic sign, and 2V. Significant improvement of this traditional method will require a technical breakthrough.

SEM method: Minerals in thin sections can also be identified based on their chemical compositions. Automated SEM provides an alternative means of mineral identification. SEM uses a focused electron beam to scan the material and generate an image of the grains. The interaction of the electrons with atoms on the grain surface provides additional information captured by the various sensors to determine the chemical composition of the mineral. SEM output includes the chemical composition with grain size, shape, and proportions. Grain mounting can be performed using an electron microprobe. However, this method is time consuming.

In our approach, we will make elemental maps of thin sections using SEM and/or E-probe, so that each mineral grain can be unambiguously identified. The same set of thin sections will be imaged using PP light, XP light, and reflected (R) light under a petrographic microscope. Our well-trained personnel will use elemental maps produced by SEM/e-probe to derive mineral identity annotations for the petrographic images, which will be used to train machine learning models to identify minerals using petrographic images. To build the training data, we anticipate collecting the SEM elemental maps of 100 Apollo lunar thin sections, which will be annotated for mineral identities by our well-trained personnel. Also, We will obtain the petrographic images (PPL, XPL, RL) images on these 100 Apollo lunar thin sections, and associate them with the mineral identities obtained by using SEM.

Developing ML-based computer vision techniques for identifying minerals in Apollo lunar thin sections: Machine learning (ML)-based methods are increasingly applied in a multitude of fields, including automated driving and navigation, automated image recognition, automated medical

diagnosis, and agricultural processes [2]. ML tools and methods also extend to Earth and Planetary Sciences. We will train a semantic segmentation neural network to identify minerals. A comprehensive dataset consisting of PP, XP, and R light microscopy images is needed to train a segmentation neural network. Each type of imaging provides unique information about the mineral's optical properties. PP light images reveal color and pleochroism, XP light images display interference colors and optical orientations, and R light images highlight surface reflections and textures. We will annotate the images with exact mineral types for training. We will develop a deep neural network with multiple input streams that merge before the final segmentation layers. As the training samples are limited, the team will utilize a backbone network, such as ResNet, as the mineral identification base network, and fine-tune it with the newly collected mineral dataset. The training process involves feeding the network with the collected images and their corresponding annotations. The objective is to minimize the difference between the network's output and the ground truth annotations. This is typically done using loss functions such as cross-entropy for categorical segmentation tasks. Moreover, incorporating techniques like data augmentation (e.g., rotations, scaling, and color adjustments) can enhance the model's ability to generalize from the training data to unseen data. Regularization methods such as dropout or L2 regularization should be used to prevent overfitting.

Analyzing images captured at different rotation angles in PP and XP light is essential for comprehensive mineral identification using microscopy. Under PP light, rotation helps in observing changes in color and light absorption, critical for identifying pleochroism in anisotropic minerals. Under XP light, rotation is vital to discern birefringence patterns, as minerals may exhibit different interference colors or go extinct at specific angles due to their crystallographic orientation. An effective strategy involves an early fusion approach where features from these different branches are combined at an early stage in the network to exploit the interdependencies between angles. This can help the network better understand the complex optical properties of minerals, which are critical for accurate classification. By training on this labeled dataset, the network can develop a robust model capable of recognizing subtle differences in mineral properties that are only visible under specific lighting and angular conditions.

We will fuse the images in different modalities to enhance the mineral recognition effects. This fusion can be achieved through various methods such as concatenation or feature-level fusion at different stages of the network. Each input stream processes one type of image (PP, XP, or R light), and their features are combined to exploit the complementary information provided by each modality. The fusion is based on higher-level features extracted by the initial layers of the network. This can be achieved by concatenating the feature maps from separate streams dedicated to each modality. The combined feature maps then continue through subsequent layers of the network. This method is particularly beneficial for capturing complex interactions between features extracted from different types of images. We will develop blending techniques such as additive or multiplicative blending that can be used to merge features. Element-wise addition or multiplication of feature maps can support the network to learn more complex representations of the merged data. As different image modalities contribute differently to

segmentation, attention mechanisms will be integrated to allow the network to focus on more informative parts of the fused features. Channel-wise or spatial attention modules can dynamically weigh the importance of features from different modalities during the training process. The effective fusion of multiple image modalities at different rotation angles into a neural network can significantly enhance the accuracy of mineral identification systems.

As another sub-track, we will also develop innovative AI/ML methods to tackle accuracy and uncertainty in 3D chemical mapping during 3D reconstructions of Apollo lunar soil samples. Our approach is to use multi-view photogrammetry with SEM images of Apollo lunar soil samples. We will use a collection of SEM images acquired from multiple viewpoints to build a 3D point cloud gathering millions of surface points with AI/ML algorithms. Many geologists use a commercial package, Agisoft's metashape, to create 3D models for 3D outcrops modeling; however, we will develop our own algorithms and we will evaluate our method with all others, including the open-source softwares and commercial packages. Based on that, we will develop innovative AI/ML methods for tackling accuracy and uncertainty during 3D reconstructions.

2.3 Broader Impact

Our ultimate goal is to develop an automatic mineral identification system for thin sections of terrestrial rocks and extra-terrestrial rocks through petrographic images and innovative AI methods in 3D chemical mapping of lunar samples. At the proof-of-concept stage, we will focus on lunar samples, because:

- a) The number of minerals identified in lunar rocks is much smaller than that on Earth. This is because the Moon is essentially dry, which prevents a lot of minerals from forming.
- b) We expect to introduce NASA's Apollo missions and NASA's PDS data to high school students and the public at large. We will teach high school students how to identify minerals in Apollo thin sections manually; we will also involve high school students in our cross-disciplinary research integrating planetary science and AI. Ping Wang is currently running an NSF-funded 3-year, \$2M, AI outreach project by which we will teach high school students how to segment and annotate mineral grains in Apollo thin sections through this project.

2.4 Labs, Samples and Instruments to Use

- a) Apollo lunar thin sections: We will work with JSC Lunar Curator Dr. Ryan Ziegler to identify about 100 thin sections for this proof-of-concept study. Note that, these thin sections would never leave ARES/JSC during this proposed study.
- b) Apollo lunar soil samples: We will work with JSC lunar Curator Dr. Ryan Ziegler to identify about 10 soil samples for this proof-of-concept study. Note that these soil samples would never leave ARES/JSC during this proposed study.
- c) Keyence automated microscope with imaging capability
- d) SEM

2.5 Samples, Procedures, and Time Frame

Because large data is needed to develop an AI/ML-based method, we plan to work on 100 Apollo lunar thin sections and 10 Apollo lunar soil samples. At the proof-of-concept stage, we

propose to start with 10 thin sections. We anticipate finishing all the work starting 2025 and before the end of 2026.

- a) For each thin section, we will use a petrographic microscope to examine 40-50 circular fields of view with a radius of 1 mm. At each field of view, we will produce 72 PP light, 72 XP light, and 1 reflective image.
- b) For the same thin section, we will use SEM to produce elemental maps for each thin section.
- c) For the soil samples, we will use SEM to produce BSE images for each soil sample.

2.6 Previous Experience

Ping Wang is a Research Associate of the Earth, Environmental, and Planetary Sciences Department, University of Tennessee, Knoxville. She has extensive experience in working with rocks samples, including thin sections of terrestrial rocks, meteorites, and Apollo lunar rocks.

- a) She worked at Harvard during 2008-2014 with Dr. Stein Jacobsen on thin sections of meteorites and Apollo lunar thin sections
- b) She worked at University of Nevada, Las Vegas during 2014-2021 with Dr. Shichun Huang on thin sections of meteorites and terrestrial rocks from Hawaii Scientific Drilling Project.
- c) She is working at University of Tennessee, Knoxville 2023- present: as the Leading PI of NSF-funded Planet+AI outreach project integrating planetary science and AI.
- d) She has been collaborating with **Dr. Guoyu Lu**, an Assistant Professor of the School of Electrical and Computer Engineering at University of Georgia, on AI4Science projects. Lu has extensive experience in developing AI systems on various applications, such as scene segmentation and analysis, scene understanding, 3D geo-reconstruction and registration research for autonomous driving.

Shichun Huang is the Sisk Associate Professor of the Earth, Environmental, and Planetary Sciences Department, University of Tennessee, Knoxville. He is an igneous petrologist, with extensive experience in petrologic and petrographic studies of both terrestrial and extraterrestrial rocks.

3 Sample Issues

We request the following Apollo lunar thin sections for the proposed proof-of-concept work:

10003, 10009, 10017, 10018, 10019, 10020, 10021, 10022, 10023, 10024, 10027, 10032, 10044, 10045, 10046, 10047, 10048, 10049, 10050, 10056, 10057, 10058, 10059, 10060, 10061, 10062, 10063, 10064, 10065, 10066, 10067, 10068, 10069, 10070, 10071, 10072, 10073, 10074, 10075, 10082, 12002, 12004, 12005, 12006, 12007, 12009, 12010, 12011, 12012, 12014, 12015, 12016, 12017, 12018, 12019, 12020, 12021, 12022, 12031, 12034, 12035, 12036, 12038, 12040, 12043, 12045, 12047, 12051, 12052, 12053, 12054, 12055, 12062, 12063, 12064, 12065, 12072, 12073, 12075, 12076, 14006, 14035, 14041, 14042, 14045, 14047, 14049, 14051, 14053, 14055, 70002, 70017, 70019, 70135, 70136, 70148, 70161, 70175, 70181, 70185

4 Data Management

Types of data produced: Petrographic images and SEM images will be the main data for this project. We anticipate that about 30-40 thousand images will be captured. The data will be recorded by spreadsheet and be provided to the public for research via publications. Outreach and educational activities will generate digital presentations and peer-reviewed journal articles over the course of the project. Raw image data will be visually selected to exclude poor quality data, such as images captured under too strong or too weak illumination. We will also include metadata to explain experimental conditions, and data processing steps.

Policies for access and sharing: Project personnel have agreed to freely share project data within the project. The sponsor will have access to summarized data and raw data (if required) used to create annual reports.

Intellectual Property: The institutions involved in this project all maintain Intellectual Property (IP) policies. The IP guidelines of these institutions are consistent with NASA and USF policy and the Bayh-Dole Act. Ownership of IP is tied to the inventorship, as defined under US patent law, and the location of the invention. Any invention made solely at a single institution is the property of that institution. Similarly, any jointly made invention will be jointly owned by the various inventors. Regardless of ownership, parties will manage respective IP in such a way as to accomplish the objectives of this proposal.

Policies for re-use and redistribution: Although data will be publicly available at the end of the project, access to summarized project data during the project will be available for educational, research, and non-profit purposes upon request to the project directors.

Plans for archiving & preservation: Raw and processed data will remain in possession of the project investigators. They will store raw and processed data in both non-digital and digital form (e.g., local hard-drive, USB external hard-drive, or SD card), online servers, and will ensure that it is duplicated and backed up in at least one on-site and one off-site location. Image data, algorithm documents, deep neural network code and models will also be maintained on GitHub for public access. Data derived from the project will be stored on both online servers and local computers, and documented in publications and reports. Raw and processed data will remain in the possession of the investigators creating the data. We will create a metadata document to explain how the data are captured and categorized. We also will provide a list of papers published to provide references for other potential users.

Acknowledging NFAR: Our future publications that result from data obtained from the NFAR laboratories will acknowledge the facility award in the acknowledgement section.

5 References

[1] W. S. MacKenzie. Optical mineralogy, principles and practice by c. d. gribble and a. j. hall. ucl press, London, 1992. no. of pages: 302. price: £50.00 (hardback); £17.95 paperback.
Geological Journal, 30(1):80–80, 1995.

[2] Guoyu Lu, Sheng Li, Gengchen Mai, Jin Sun, Dajiang Zhu, Lilong Chai, Haijian Sun, Xianqiao Wang, Haixing Dai, Ninghao Liu, et al. Agi for agriculture. arXiv preprint arXiv:2304.06136, 2023.